

Random Modeling

2nd Year Engineering : Applied Mathematics and Modeling Engineering
Written By :Dhia Gdara

Central Limit Theorem (CLT) — Exercises with corrections

Reminder: Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with:

$$\mathbb{E}(X_i) = \mu \quad \text{and} \quad \mathbb{V}(X_i) = \sigma^2, \quad \text{for } i = 1, 2, \dots, n.$$

Then, for n sufficiently large, the standardized random variable:

$$Z = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

follows approximately the standard normal distribution $\mathcal{N}(0, 1)$.

From this, we obtain two key approximations:

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2) \quad \text{and} \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Case	Minimum n
Distribution close to normal	$n \geq 4$
Moderately close (e.g., uniform)	$n \geq 12$
Very asymmetric (e.g., exponential)	$n \geq 100$

Exercise 1 — Sum of Insurance Claims (Gamma Distribution)

Statement

An insurance company receives claims from policyholders. The amount of each claim X_i (in thousands of euros) follows a Gamma distribution with shape parameter $k = 2$ and rate parameter $\theta = 0.5$. For this distribution:

$$\mathbb{E}(X_i) = \frac{k}{\theta} = \frac{2}{0.5} = 4, \quad \mathbb{V}(X_i) = \frac{k}{\theta^2} = \frac{2}{(0.5)^2} = 8.$$

The company receives $n = 100$ independent claims. Let $S_{100} = X_1 + X_2 + \cdots + X_{100}$ be the total amount of claims.

- Justify the application of the CLT for $n = 100$.
- Give the approximate distribution of S_{100} .
- Calculate $\mathbb{P}(S_{100} > 420)$.
- Calculate $\mathbb{P}(380 \leq S_{100} \leq 430)$.

Detailed Correction

a. Justification of the CLT The Gamma distribution (with $k = 2$) is moderately asymmetric. According to the CLT guidelines:

- If the distribution is close to normal: $n \geq 4$
- If moderately close: $n \geq 12$
- If very asymmetric: $n \geq 100$

Here the Gamma distribution is not extremely asymmetric, but to be safe and accurate, $n = 100$ is considered sufficiently large. The CLT applies.

b. Approximate distribution of S_{100}

$$\mathbb{E}[S_{100}] = n\mu = 100 \times 4 = 400$$

$$\mathbb{V}[S_{100}] = n\sigma^2 = 100 \times 8 = 800$$

$$\sigma_S = \sqrt{800} = 20\sqrt{2} \approx 28.284$$

$$\boxed{S_{100} \approx \mathcal{N}(400, 800)}.$$

c. Calculation of $\mathbb{P}(S_{100} > 420)$ Standardize using $Z = \frac{S_{100} - 400}{\sqrt{800}}$:

$$\mathbb{P}(S_{100} > 420) = \mathbb{P}\left(Z > \frac{420 - 400}{28.284}\right) = \mathbb{P}\left(Z > \frac{20}{28.284}\right) = \mathbb{P}(Z > 0.7071)$$

From the standard normal table: $\Phi(0.7071) \approx 0.7602$

$$\mathbb{P}(Z > 0.7071) = 1 - 0.7602 = 0.2398$$

$$\boxed{\mathbb{P}(S_{100} > 420) \approx 0.240}.$$

There is approximately 24% chance that total claims exceed 420,000 euros.

d. Calculation of $\mathbb{P}(380 \leq S_{100} \leq 430)$

$$z_1 = \frac{380 - 400}{28.284} = \frac{-20}{28.284} \approx -0.7071$$

$$z_2 = \frac{430 - 400}{28.284} = \frac{30}{28.284} \approx 1.0607$$

$$\begin{aligned} \mathbb{P}(380 \leq S_{100} \leq 430) &= \mathbb{P}(-0.7071 \leq Z \leq 1.0607) \\ &= \Phi(1.0607) - \Phi(-0.7071) \\ &= \Phi(1.0607) - [1 - \Phi(0.7071)] \end{aligned}$$

From tables: $\Phi(1.0607) \approx 0.8554$, $\Phi(0.7071) \approx 0.7602$

$$= 0.8554 - (1 - 0.7602) = 0.8554 - 0.2398 = 0.6156$$

$$\boxed{\mathbb{P}(380 \leq S_{100} \leq 430) \approx 0.616}.$$

Exercise 2 — Average Waiting Time in a Queue (Exponential Distribution)

Statement

In a call center, the waiting time X_i (in minutes) for each customer follows an exponential distribution with parameter $\lambda = 0.2$. For an exponential distribution:

$$\mathbb{E}(X_i) = \frac{1}{\lambda} = \frac{1}{0.2} = 5, \quad \mathbb{V}(X_i) = \frac{1}{\lambda^2} = \frac{1}{(0.2)^2} = 25.$$

A sample of $n = 50$ independent customers is observed. Let $\bar{X}_{50}$ be the average waiting time.

- Justify the application of the CLT for $n = 50$.
- Give the approximate distribution of $\bar{X}_{50}$.
- Calculate $\mathbb{P}(\bar{X}_{50} > 6)$.
- Find the value c such that $\mathbb{P}(|\bar{X}_{50} - 5| \leq c) = 0.95$.

Detailed Correction

a. Justification of the CLT The exponential distribution is very asymmetric (it has a long right tail). According to the CLT guidelines:

- Very asymmetric distributions (exponential, Pareto, etc.) require $n \geq 100$ for excellent approximation.

Here $n = 50$ is less than 100, but the CLT still provides a reasonable approximation (many textbooks accept $n \geq 30$ for moderate accuracy). We will apply it with caution.

b. Approximate distribution of $\bar{X}_{50}$ For the sample mean:

$$\mathbb{E}[\bar{X}_{50}] = \mu = 5$$

$$\mathbb{V}[\bar{X}_{50}] = \frac{\sigma^2}{n} = \frac{25}{50} = 0.5$$

$$\sigma_{\bar{X}} = \sqrt{0.5} \approx 0.7071$$

$$\boxed{\bar{X}_{50} \approx \mathcal{N}(5, 0.5)}.$$

c. Calculation of $\mathbb{P}(\bar{X}_{50} > 6)$ Standardize using $Z = \frac{\bar{X}_{50} - 5}{\sqrt{0.5}} = \frac{\bar{X}_{50} - 5}{0.7071}$:

$$\mathbb{P}(\bar{X}_{50} > 6) = \mathbb{P}\left(Z > \frac{6 - 5}{0.7071}\right) = \mathbb{P}\left(Z > \frac{1}{0.7071}\right) = \mathbb{P}(Z > 1.4142)$$

From the standard normal table: $\Phi(1.4142) \approx 0.9213$

$$\mathbb{P}(Z > 1.4142) = 1 - 0.9213 = 0.0787$$

$$\boxed{\mathbb{P}(\bar{X}_{50} > 6) \approx 0.0787}.$$

There is approximately 7.9% chance that the average waiting time exceeds 6 minutes.

d. Finding c such that $\mathbb{P}(|\bar{X}_{50} - 5| \leq c) = 0.95$ We have $Z = \frac{\bar{X}_{50} - 5}{0.7071} \sim \mathcal{N}(0, 1)$. The condition becomes:

$$\mathbb{P}(|Z| \leq c^*) = 0.95 \quad \text{where} \quad c^* = \frac{c}{0.7071}$$

We know that for a standard normal:

$$\mathbb{P}(|Z| \leq 1.96) = 0.95$$

Thus:

$$\frac{c}{0.7071} = 1.96 \quad \Rightarrow \quad c = 1.96 \times 0.7071 \approx 1.386$$

$$\boxed{c \approx 1.386 \text{ minutes}}.$$

Interpretation: With 95% probability, the sample mean waiting time $\bar{X}_{50}$ lies within ± 1.386 minutes of the true mean $\mu = 5$ minutes. That is:

$$\mathbb{P}(3.614 \leq \bar{X}_{50} \leq 6.386) \approx 0.95.$$